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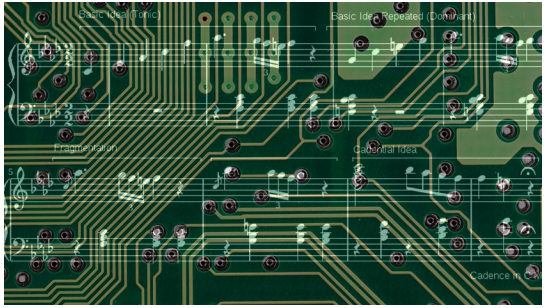
MIR in practice – Ethics, Responsible Research, and Bias

Dr Eamonn Bell

COMP3721 Introduction to Music Computing

Case A

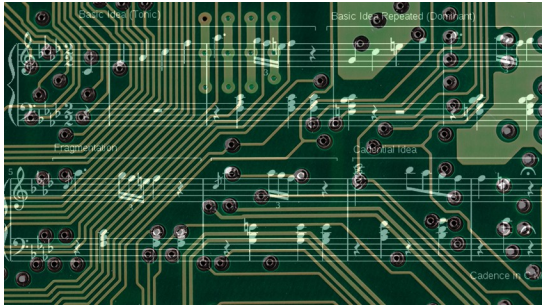
“Related to their work supported by a public research grant, Adaetal wonder how well a particular machine learning method can be used for music modelling and generation in a specific style, given machine-readable notations. Adaetal know about the website httpabcmusic, which has tens of thousands of transcriptions of traditional music contributed by hundreds of people who play the music. Adaetal download all of the transcriptions and use the machine learning method to build a model of the collection.



[Holzapfel et al. 2016]

Case A

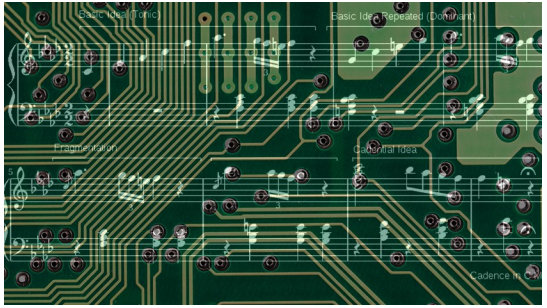
“Adaetal wonder how well the resulting model has captured the stylistic conventions of the music, and so they use the model to generate thousands of new transcriptions, synthesise them using a variety of stylistically appropriate MIDI-controlled instruments, and post them online for anyone to hear. Adaetal return to httpabcmusic and ask its users in a discussion forum to listen to the results and say what they think. Some users are amused, some are not impressed, and some are offended.



[Holzapfel et al. 2016]

Case A

“To try to encourage a more substantial discussion about the quality of the computer-generated transcriptions, Adaetal selects 3,000 of them at random to create a volume of transcriptions. Adaetal then hires music experts to look over the volume, and perform some at concerts. One of the musicians Adaetal are working with suggests recording and releasing a CD of the computer-generated music, but passed off as composed by a real, but unknown composer in that tradition. The CD could be reviewed by a specialist, and then revealed as being generated by a computer. Adaetal write a grant proposal to support such a project, but they have second thoughts when the idea invokes a strong rebuke from another expert in the tradition.”



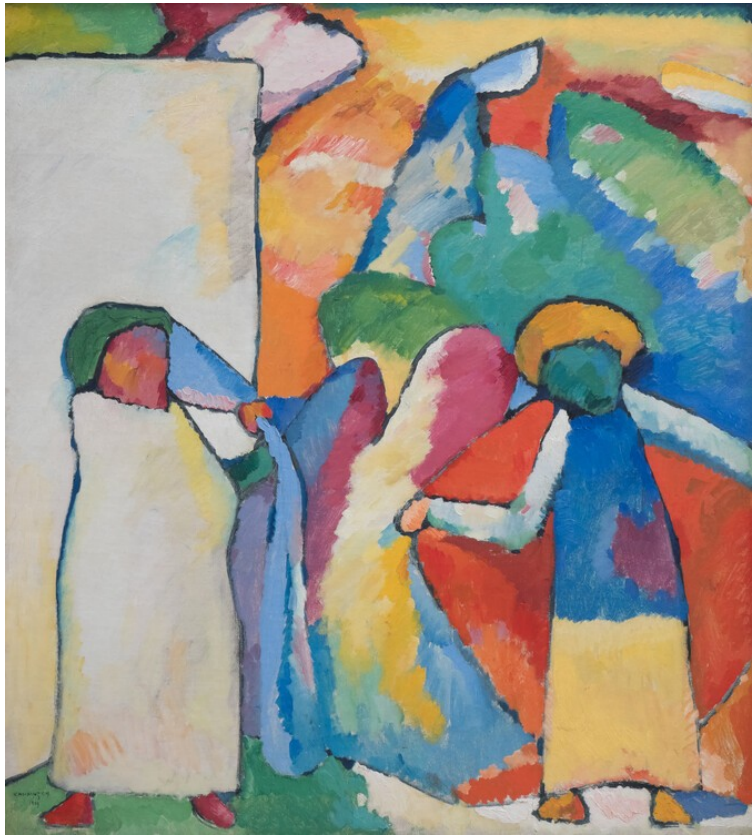
[Holzapfel et al. 2016]

Research Ethics

Ethical norms govern research, just as in every other sphere of social life

There is a special responsibility on researchers to ensure that the research they do – which is understood to add value to society – does not come at the cost of treating others poorly

Principles (e.g. University of Manchester): **autonomy, beneficience, non-maleficience, justice, confidentiality, <u>integrity</u>**



Ethical Approval

Research projects benefit from early engagement with others, and most institutions have a process to facilitate this:

- Research Ethics Committee
- Institutional Review Board (IRB)
- Ethics Review Board (ERB)

Projects involving humans or data about humans are likely to fall in the scope of these bodies

Sometimes, your project will be exempt. But if there is any doubt, seek institutional ethical approval – it is formative!

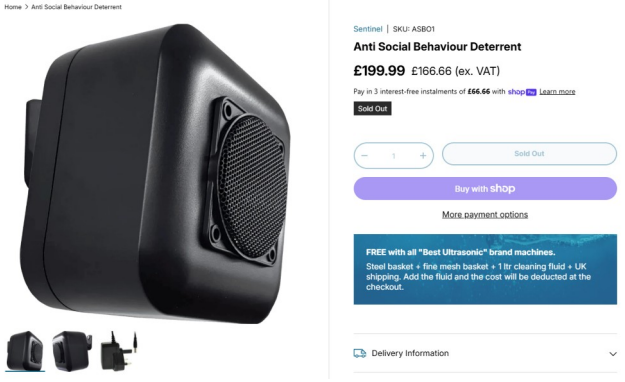
Typical responsibilities of an ethics review process

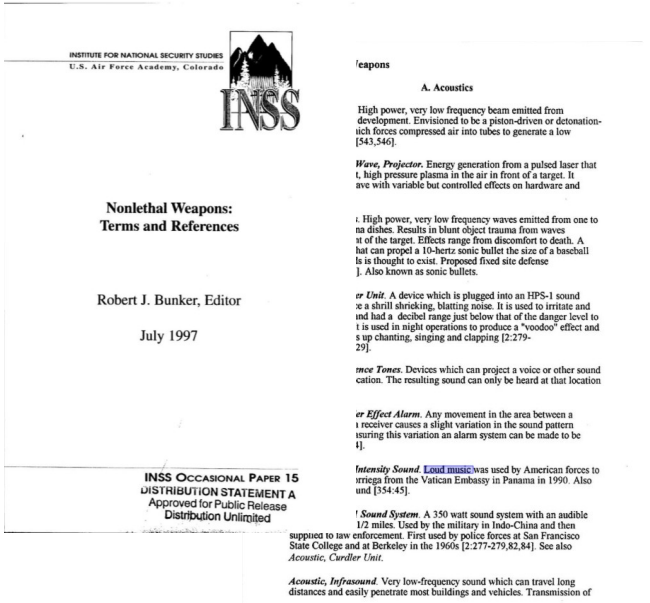
- maintaining ethical standards of practice in research with potential to do harm
- protecting human participants from potential harm
- protecting researchers from potential harm
- the preservation of participants' dignity, rights, and welfare
- taking account of legitimate interests of other individuals, bodies, and communities associated with the research and providing reassurance to the public and to outside bodies that their legitimate interests have been protected
- operating free from perceived or actual conflicts of interest so that the application of ethics principles and reasoning is neither impeded nor compromised

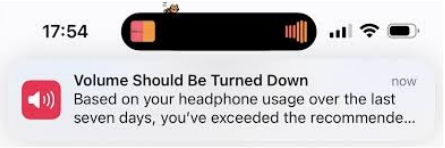
[https://www.bath.ac.uk/guides/undergraduate-and-masters-research-projects-ethics-review/]

What about music?

Public Domain (WPA) c. 1940







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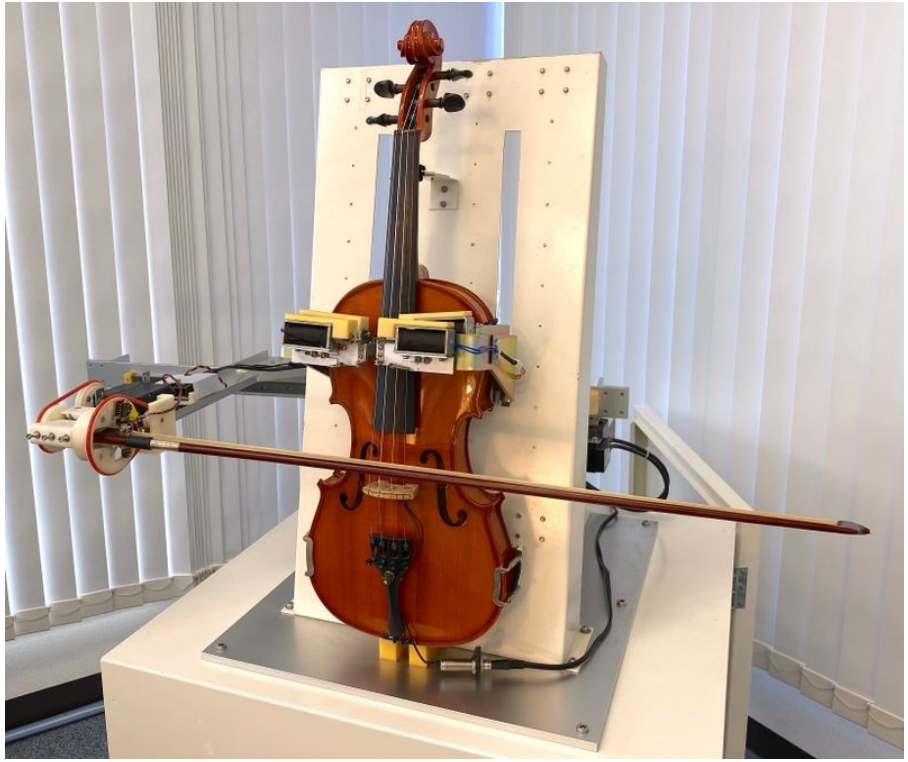
What about MIR?



[Kamatani et al. 2022]

Responsible Research and Innovation

Responsible Research and Innovation (**RRI**) encompasses the wider ethical and societal effects of research and innovation activity

RRI is not only in the purview of academic research

Example: '"popularity bias", a supposed tendency for music recommender systems [MRS] to favour items that are already popular, thus reinforcing or amplifying the success of the most successful artists and companies, which might also potentially limit the access of audiences to a wider diversity of music' [Hesmondhalgh et al. 2023]

Responsible Research and Innovation

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[Fig. 2, Holzapfel et al. 2016]

A framework for RRI: AREA

A – Anticipate

Based on existing evidence, describe and analyse intended and unintended impacts of what might arise

R – Reflect

Reflect on the purposes of, motivations for and potential implications of the research, including dilemmas and social change

E – Engage

Open up to broader deliberation, dialogue, engagement and debate

A – Act

The results influence the direction and trajectory of the research and innovation process itself (including commercialisation!)

[Owen et al. 2013]

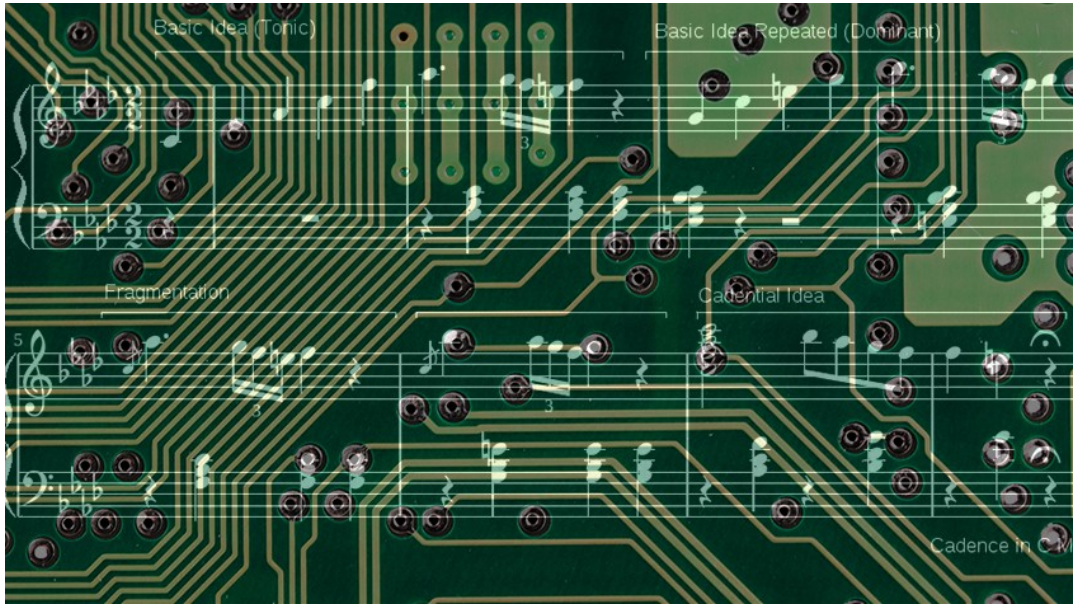
Back to Case A

A – Anticipate

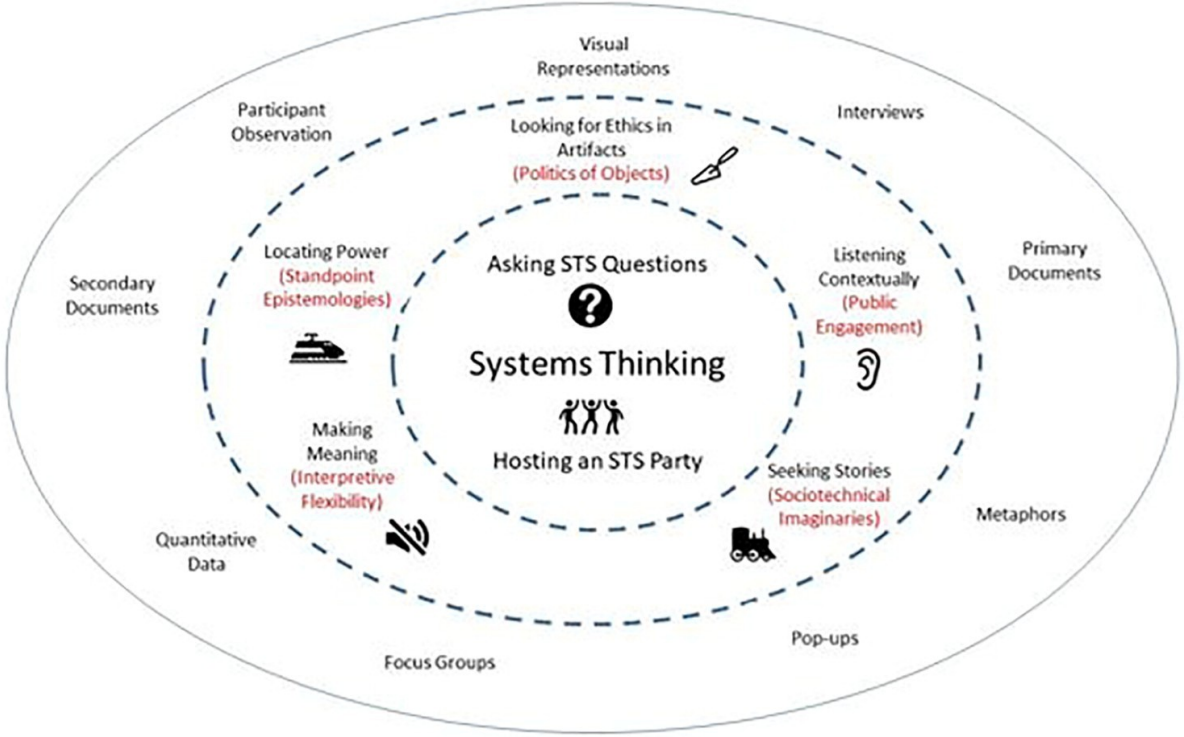
R – Reflect

E – Engage

A – Act



Approaches from Science and Technology Studies (STS)



[Fig 1., Tomblin and Mogul 2020]

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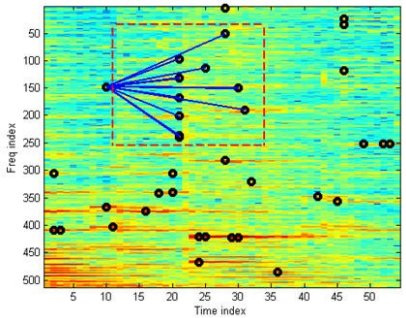
Case B

“Systems for music detection and recognition can be applied to count the number of hours that radio stations broadcast music, and/or to identify the recordings played. The government of Lalaland decides to apply a cover song detection system to all radio broadcasts to prevent the illegal reproduction of copyright-protected music. Radio broadcasters are obliged to pay amounts according to the time measured by the installed systems, but traditional Lalalandian tunes that do not underlie any copyright restrictions get a large part of the airplay in Lalaland. However, the system incorrectly attributes most of the indigenous music as cover versions of Beatles songs. Broadcasters thus reduce the amount of time they broadcast local (and other) music, in order to reduce their costs



[Roger Jang]

[Holzapfel et al. 2016]

Another lens: “bias”

Recommendation systems (among others) are often biased – i.e. they systematically and unfairly discriminate against certain individuals or groups

Three types of bias:

1. **Pre-existing** bias (e.g. demographic and sociocultural of research producers)

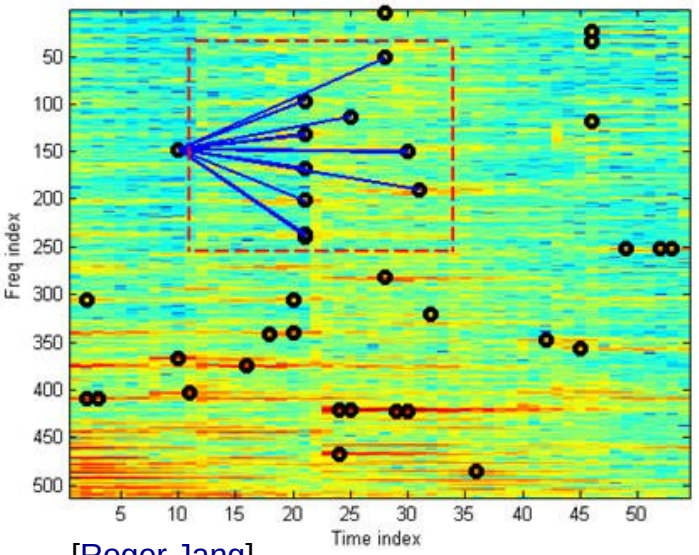
2. **Technical** bias (e.g. datasets overrepresent “Western” music, shared tasks – e.g. competitions and benchmarks serve these idioms, as do metrics)

3. **Emergent** bias (e.g. different countries organise music industry differently, flawed knowledge/models persist due to inertia, even in “just” systems)

[Holzapfel et al. 2016]

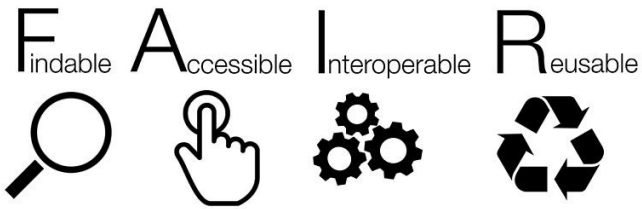
Back to Case B

- **pre-existing** bias
- **technical** bias
- **emergent** bias

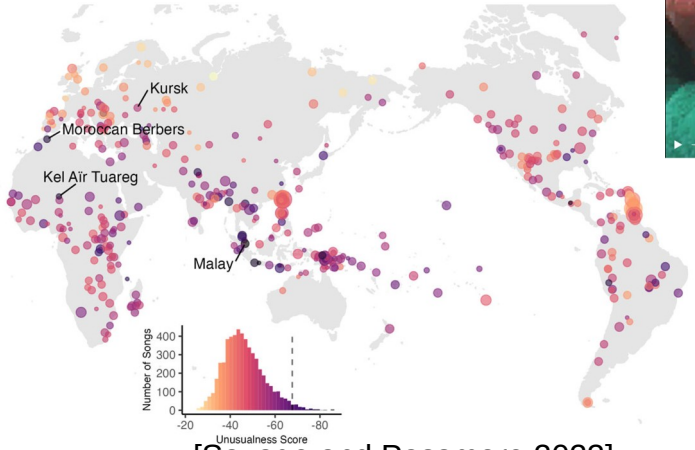


[Roger Jang]

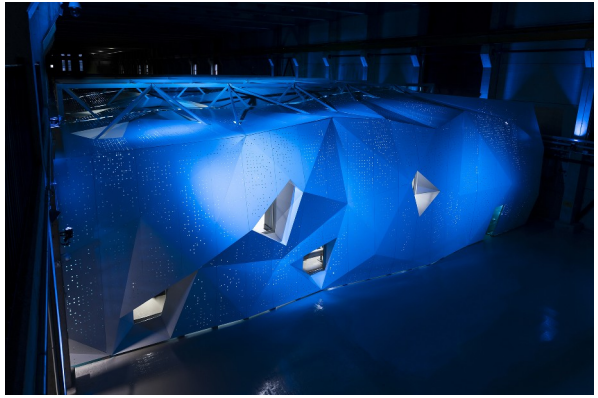
Attributes of a responsible MIR R&I ecosystem?



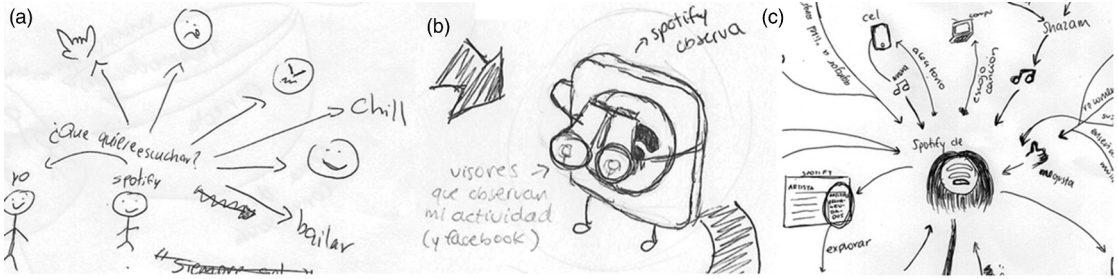
CC-BY-SA 4.0 SangyaPundir



[Savage and Passmore 2023]



LUMI [Credit: Fade Creative]



[Siles et al. 2020]